

7. Write a C Program to swap or interchange two integers using call by reference method.

```
#include <stdio.h>
```

```
// Function Prototype using pointers
```

```
void swap(int *p1, int *p2);
```

```
int main(void) {
```

```
    int a, b;
```

```
    printf("Enter two integers (a and b): ");
```

```
    scanf("%d %d", &a, &b);
```

```
    printf("\nBefore Swapping: a = %d, b = %d", a, b);
```

```
// Function call: passing the addresses of  
a and b
```

```
swap(&a, &b);
```

```
printf("\nAfter Swapping: a = %d, b =  
%d\n", a, b);
```

```
return 0;
```

```
}
```

```
// Function definition to swap values
```

```
void swap(int *p1, int *p2) {
```

```
    int temp;
```

```
    // Logic using a temporary variable
```

```
    temp = *p1; // Store value at address p1  
    in temp
```

```
*p1 = *p2; // Put value at address p2 into  
address p1
```

```
*p2 = temp; // Put temp value into  
address p2
```

```
}
```